A larger Paleolithic toolkit

27 new item types, actual materials and usable camp equipment. Prototype art; functional acceptance is tracked separately.

The pass follows the agreed order: regional sapiens audit, Paleolithic content, crowded NPC performance, then rivers. These pages distinguish tested gameplay from the work still ahead.



Actual in-game spear: warmed compound adhesive on the haft and measured ochre markings.

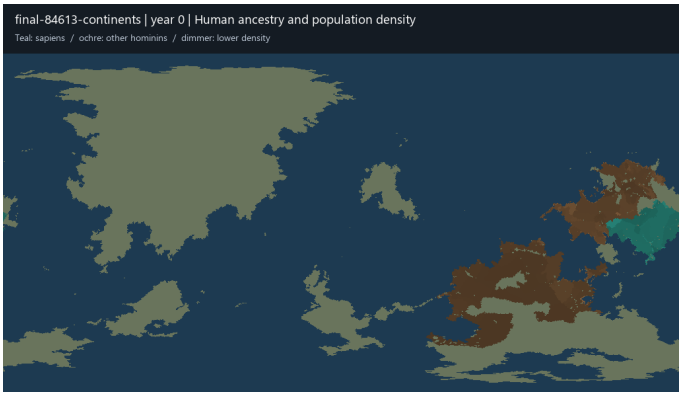
New family	What it adds
Clothing and carrying water	Six wearable garments; 2 L waterskin and 1.5 L bark vessel.
Food, pottery and stonework	Grinding surface, meal and cakes; gathered clay, drying, firing and 3 L pots; prepared cores and pressure finishing.
Campcraft and materials	Reed mats, bone pendants, gum/resin adhesives, pigments and charcoal.
Travel and trapping	Finite-fuel torches and ember carriers; real-material deadfalls; reed rafts, hide-frame boats and carved paddles.

Craft through the existing item context actions, using carried or nearby materials. Previous hands are borrowed and restored through the existing work system. No in-game recipe book or starter bag was added.

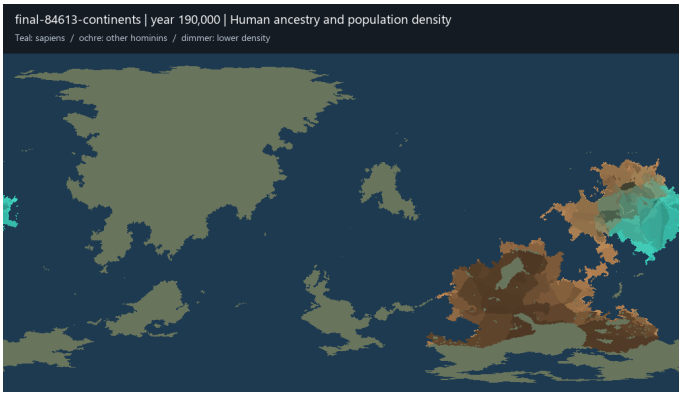
Different worlds, bounded dispersal

Four fresh 200,000-cell maps; elapsed years 0 to 190,000. These are data-audit maps, not new geography.

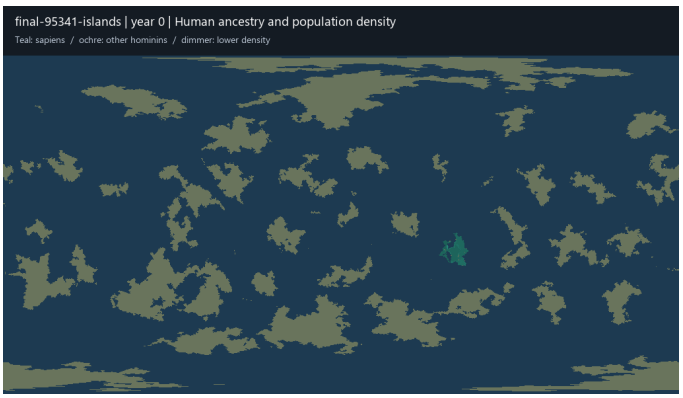
Pangeae, Iceage, Continents and Archipelago were checked at seven dates. Regional climate, carrying capacity, technology and water crossings constrain expansion; distant regions do not become one uniform population.



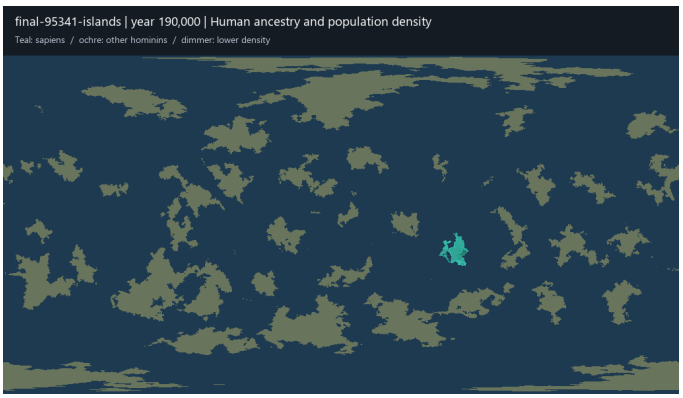
Continents seed 84613, starting distribution.



Same world after 190,000 elapsed years. Teal: sapiens; ochre: other hominins.



Archipelago seed 95341, starting distribution.



Separated landmasses retain distinct occupation histories.

Specific fix: when carrying capacity falls to zero, analytic projection now retains the calculated cold/starvation mortality instead of replacing it with a gentler fixed decline. The simulation still works on regional aggregates; local camps and people are generated when needed.

Evidence: 52 mechanism checks and 12 temporal/save/cancellation checks. The targeted zero-capacity test changes correctly; two paired sampled world totals remained identical. This is a focused correction and audit, not a claim that every migration rate was retuned.

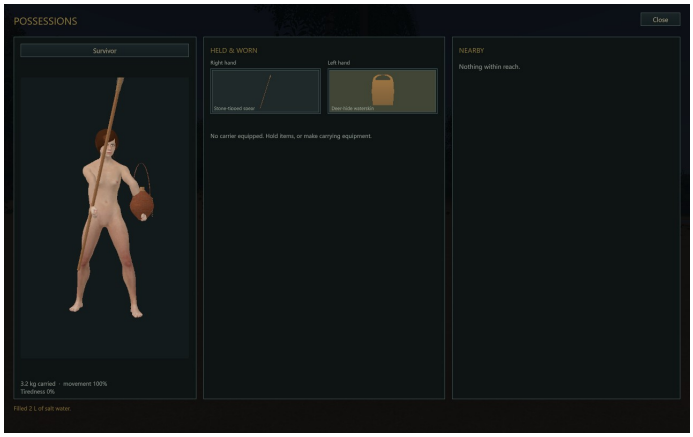
Sources: Nature: niche expansion · Nature: Arabian occupation windows

Clothes and portable water

Owned, worn and filled possessions with real mass and material provenance.



Garments fitted to the actual character.



Finite carried water and its normal inventory context.

Equipment	Use
Hide wrap, tunic or fur coat	Independent torso attachment. Worn insulation affects exposure; packing clothes provides no protection. Heavy insulation increases hot-weather effort.
Leggings, footwear and hood	Separate legs, feet and head attachments coexist with bags and weapons. Hood handles existing hair visibility.
Waterskin / bark water vessel	2 L / 1.5 L. Fill at real water, drink a portion or pour it away. Contents add actual mass and retain salinity and contamination.

Hide equipment preserves the donor animal and coat. Local NPC outfits use their band's practiced techniques and scale to children. Garment silhouettes are a functional first pass using native body weights, not finished draped clothing art.

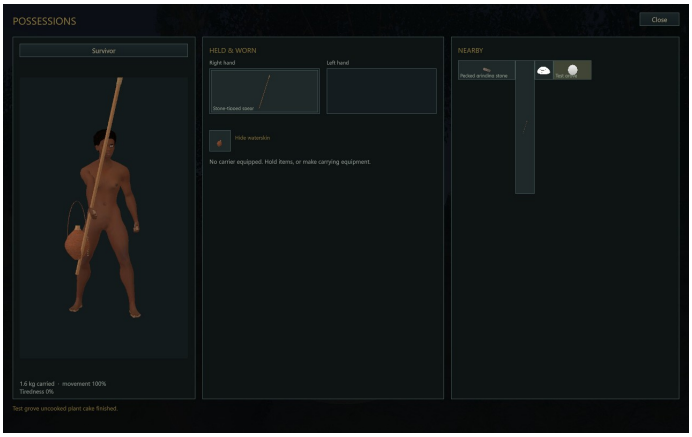
Water actions retain the player's actual hands. Boiling addresses biological contamination; it does not turn salt water fresh. Adding untreated water can contaminate a previously boiled supply.

From gathered soil to a meal

Measured ingredients, elapsed time and actual fire heat.



Fired pot with measured boiled water.



Grinding surface and reusable pounding/cutting equipment.

Progression	What happens
Clay to fired pot	Dig suitable mapped clay with a digging stick. Shape with 100 ml water, air-dry for roughly a solar day, then fire the same pot in a well-fueled hearth.
Boiling	Heat a fired pot directly. Bark/skin vessels use actual heated hearth stones, with a finite stored-heat budget.
Meal and cake	Peck an ordinary cobble into a grinding surface. Grind nuts or roots with a separate pounder. Mix meal with 80 ml carried water and cook the cake at a fire.

Mass, calories and plant toxins follow the source food. Mixing does not manufacture calories or make poisonous plants safe. Wet clay, water quality, drying, firing and cooking survive saving.

Limits: elapsed weather uses the current local conditions as an approximation; pottery and food models remain prototype art.

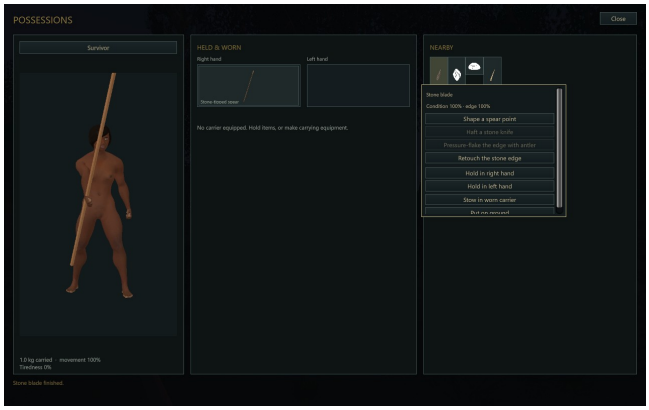
Sources: Early plant processing

Work the material, keep its identity

Prepared faces, finer edges and compound hafting.



An ordinary stone with its prepared working face.



Fine chert finishing with an actual antler retoucher.



Source pigment and adhesive carried through to the actual weapon.

Stonework: prepare a suitable ordinary core, then strike a controlled flake. Preparing and removing the flake consumes real stone. Pressure finishing currently uses fine chert; finished point quality remains through hafting.

Adhesives: collect tree gum or resin, grind ochre or recover charcoal, then warm a mixed binder on actual hearth stones. Apply five grams while workable to seal a real haft. Ochre markings use two grams and five millilitres of water.

Actual-world resource checks found suitable clay/ochre soil and generated gum/resin trees. The native glue test used supplied raw gum; it did not claim a tree-harvesting test. No ordinary stone was renamed into an abstract hammerstone or core item.

Sources: Experimental gum/ochre adhesives · Resin/charcoal experiments

A working camp, carried forward

Woven bedding, simple adornment, transported fire and physical traps.



Actual placed reed mat, after ground clearing settles.



Wearable bone pendant; its own neck attachment.



Lit torch in the real first-person view.



A carved trigger, actual stone and a small portion of food bait.

Mat: eight weaving-strip bundles and a binding become a rolled mat. Spread it as bedding, layer it for more protection, or use it as reed covering. **Pendant:** carve and pierce source bone, then string and wear it.

Fire: wrapped bark/resin torches have finite fuel. An insulated carrier takes five grams of actual hearth coals and twenty grams of tinder. Both can light a prepared fire in survival. Wind, rain, shelter, immersion and elapsed fuel are handled.

Deadfall: place a carved trigger, add a broad 2-25 kg stone and 25 g of real food, then set it. A locally simulated small animal must cross the footprint. The native rabbit was wounded and escaped; the trap did not guarantee a kill or invent a catch.

First-pass limits: approximate 90-minute torch / eight-hour ember durations; no offscreen catches, snare variants or autonomous NPC trap tending. Fire shares the existing four-light budget.

Sources: Deadfall mechanics, primary trapping manual

Launch from the shore

Reed floats and donor-hide frame boats, assembled from finite materials.



Actual constructed raft, afloat with the player.



Skin hull with the real donated deer coat texture.

Use **B** for the adjustable watercraft plans and lay them out in shallow water. Add crosspieces, reed floats or bent ribs, lashings and fitted hides; the skin boat also needs sealed seams. Hew a paddle from a genuinely thick limb using chopping and cutting tools.

E boards/leaves; **W/S** paddle and **A/D** turn. Actual hull/load mass controls draft and overloads; shores and obstacles stop travel; currents carry the active craft. Both passed actual keyboard paddling and full save/reload checks; hull supplies were finite QA inputs.

Limits: prototype models and standing rider; no authored rowing animation, capsizing or unattended voyage simulation. Unoccupied craft remain at their saved location.

Sources: Mariners' Museum: reed raft · Science Museum: hide-frame coracle

Try it in progression

External testing guide; nothing here was added as an in-game recipe book.

Start with	Action / next use
Ordinary stone and striker	Flake a cutting edge; prepare a grinding surface; later prepare a core or pressure-finish fine chert.
Fibre, reeds and hides	Prepare fibres, twist bindings/thread, split reeds; scrape and soften hide. These support clothing, water containers, mats and existing carriers.
Worked hide + binding	Cut a wrap. Cutting/piercing tools and extra hide/thread make fitted garments. Hide + thread + binding + cutting/piercing tools makes a waterskin.
Suitable clay + 100 ml water	Dig, shape, dry, fire; fill and boil. A pot holds up to 3 L.
Nuts/roots + mill + pounder	Grind meal; add 80 ml water for a cake and cook it.
Gum/ochre or resin/charcoal	Mix, warm on a hearth stone, then seal a haft. Ground ochre can mark equipment.
Pole + bark + binding + fuel	Wrap and light a torch. Bark container + bark sheet + binding + piercing tool makes an ember carrier.
Sound branch + cutting edge	Carve a deadfall trigger. Add an actual stone and a little bait after placement.
Reeds / wood / hides / bindings	Build the two watercraft through B. A paddle requires a thick limb, a chopping tool and a cutting edge.

Acceptance: All 43 technology families have a representative playable route. Final native checks cover portable crafting (38), combat (184), woodwork (16), camps (98), nature (32), land blueprints (261), fishing (14) and boats (32), alongside the new equipment checks. Functional captures are capped at 60 FPS and are not performance benchmarks. Finished art, every regional variant and autonomous NPC manufacture are outside this pass.

Remaining work: Advanced tracking, locational injuries, autonomous NPC hunting and finished item art remain later work.

Crowded camps

Stage 3 checkpoint before the river pass: 100 distinct physical NPCs, measured at eye height.



Actual in-game gathering: seven sampled bands retain their own people, personal belongings and animation graphs. This does not increase natural encounter density.

Body distance meshes and SRP-compatible materials reduce rendering work. Repeated shelter queries and inactive task checks now avoid temporary allocations. The first allocation pass measured 402 to 92 KB per recorded development-profile frame; those diagnostic timings are not release FPS.

Active crowd window	Average FPS	99th percentile	Worst frame
First	135.4	9.99 ms	20.80 ms
Repeat	136.1	9.65 ms	22.26 ms

Conditions: Medium vegetation/surfaces, 1920x1080, 6 km view, slow 10 m eye-height orbit, 30 seconds per window. All 100 residents retained; zero unfocused frames and zero unexpected render cameras. No model inference. Motors, senses, tasks, inventories and poses stay active; broad itinerary refresh is held fixed for this controlled gathering.

Limits: The strict 100 FPS floor remains unmet: occasional 20-22 ms frames remain. Other biomes and future city crowds are not certified by this scene. Automatic animation scheduling and collector-budget experiments remain disabled in normal play.

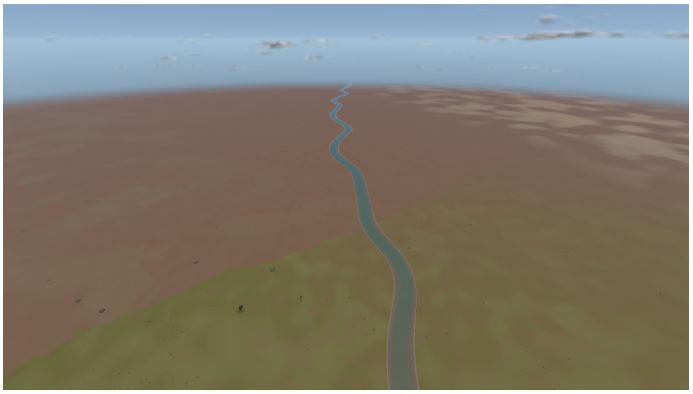
Validation: 421 exact camp-part selections and 3,877 shelter-cache controls pass. Native camp work (42), belongings (19), daily routines (33), real material use (19), combat/items (184), construction (261) and current animation controls (86) pass.

Rivers at a physical scale

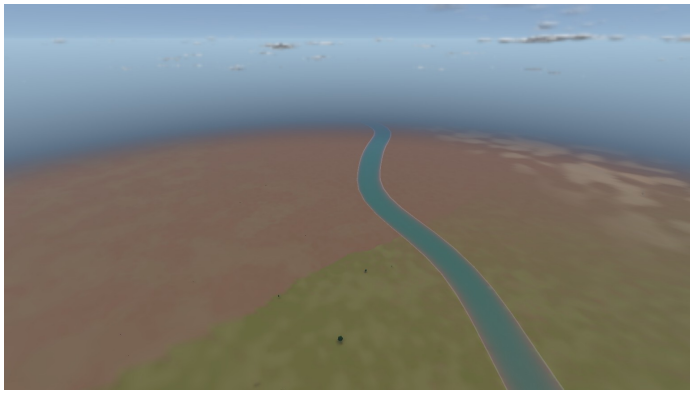
Source-globe discharge determines size. World compression controls admission, not the dimensions of an admitted river.

Width, depth and flow now come from the source river rather than compressed graph-edge limits. Irregular physical-distance profiles vary the width and pool/riffle depth; terrain confinement limits bends. Endpoint tapers preserve shared joins.

Same drainage, revised river profile



Before: source 143549, narrower channel with frequent short bends.



After 117: the same source, broader bends and a wider channel.



Before: alluvial source 115180. Existing alluvial banks were retained.



After 117: the same drainage, with revised width/depth profiles and bank fitting.

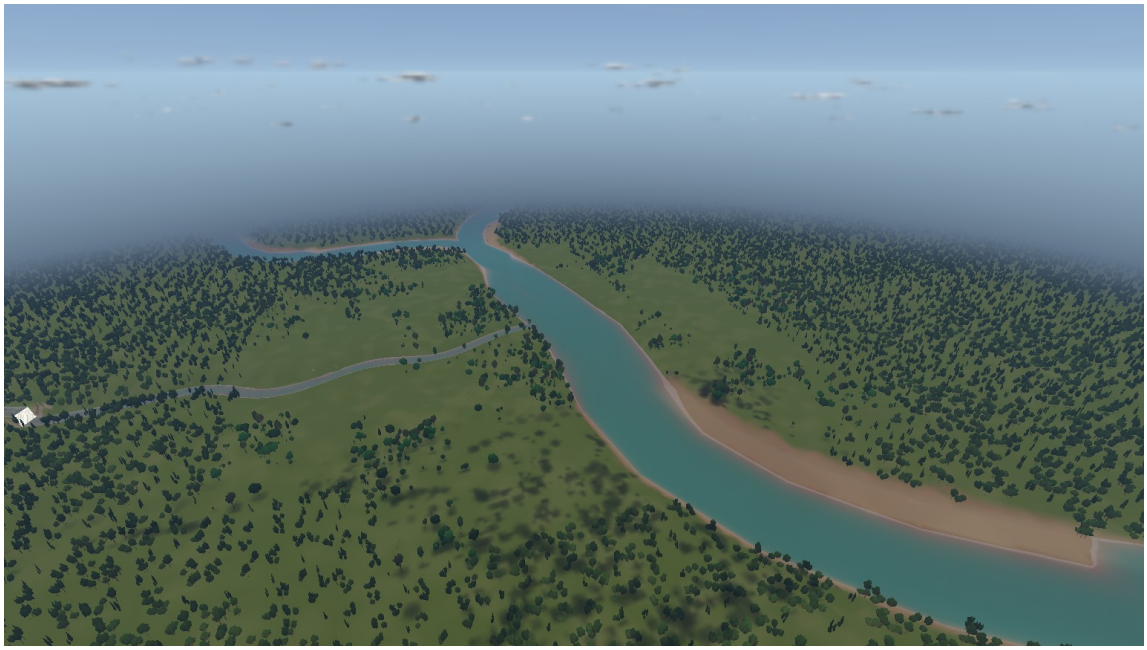
These are actual game captures at medium vegetation. The bank camera is reselected from the changed surface, so this is a same-source comparison, not an identical-camera pixel test. Some aerial banks still look angular; junction pools remain simplified.

Four deep major-river samples are 1.05-1.52 km wide and 26-34 m deep, with sampled speeds 1.57-2.03 m/s. They are examples from this world, not universal limits.

Sources: USGS hydraulic geometry · USGS channel patterns

Connected water and terrain

The final build was checked on the original failure sites. Failed intermediate trials remain recorded.



Source 52127: actual final tributaries and receiving channel. The previous bed discontinuity now passes its unchanged test.

Ordinary joins share an exact section. Confluences fit their incoming valley banks; connected lakes reconcile their water levels before fine terrain is generated. Chunk boundaries use the same shoreline height curve, and curved channel sections keep a continuous bed coordinate.

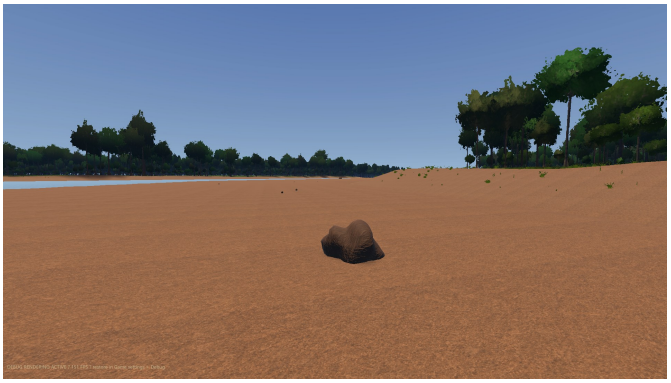
Final 117 check	Result
Two 200k-cell worlds	10,998 bank probes / 384 reaches; 0 exposed edges.
Four major sites, three LODs	1,997 water probes; 12 meshes; 0 failed probes.
Ordinary joins and chunk sea ms	384 checks at 64 nodes; mixed-LOD visible gap about 3 micrometres. 15 bank meshes:0 interior holes or folds beyond weld precision.
Lake stages and preparation order	64,636 +44,492 lake/river controls; 121,215 query-order controls pass.
Legacy 52127 curved bed	0.018 m ground change across 10 cm, below unchanged 0.1 m limit; 0 water jump.

Four native reviews pass: meandering 143549 (21 checks), alluvial 115180 (24), major 126812 (21), legacy 52127 (21). All 12 final native images were inspected. These are sampled validation, not proof of every possible world or shoreline.

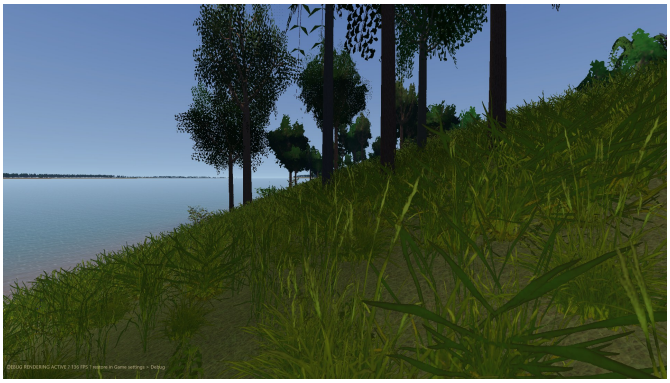
Sources: Procedural Riverscapes,2019

Moving performance and limits

Final 117 release Mono build. Ryzen 7 5700X / RTX 4060 / 1920x1080 / medium terrain and vegetation / 6 km view.



Actual alluvial-bank benchmark scene, after travel.



Actual major-river-bank scene, after travel.

30-second window	Average FPS	99th percentile	Worst frame
Alluvial / travel	239.8	11.93 ms	14.38 ms
Alluvial / turning	225.5	12.11 ms	30.15 ms
Major / travel	206.0	11.54 ms	25.12 ms
Major / turning	194.7	11.75 ms	20.85 ms

All four windows have zero unfocused frames. Travel is 8 m/s around a 30 m loop, with an additional rapid-turn case. Normal terrain, water, vegetation and animal rendering stay enabled; streaming is not held. No model inference. End-of-window streaming was still active in one case. Background-window trial results are excluded.

The strict 100 FPS floor is not met. Residual 14-30 ms frames remain, and these two banks do not certify every biome, high-speed flight or dense NPC crowds. The small instantaneous FPS overlay in the post-run stills is not the measured-window average.

Loading cost remains separate

Cold preparation took 18.60 s using the shipped Core DLL, including 16.37 s of datum fitting. The earlier original-model reference was 24.52 s total, measured in a separate window. These are isolated CoreCLR component measurements, not an end-to-end game-load comparison.

The sampled 3.28 km river tile took 1.54 s to build; a difficult 13.1 km far tile took 9.33 s in the background. Fine shoreline constraints remain expensive. Far meshes simplify bathymetry and material sampling while preserving actual outer shores. Loading is not instantaneous.

Handoff: Slot B contains 117/Hyper 104, built from 1434add9. Slot A and the normal shortcut are unchanged. Source, compact receipts and report inputs are preserved on work/river-hydraulics-20261110. River work stops here; no following geography or NPC feature was started.